

Attachment A 2

Services, Compensation, and Schedule

A. Services. Unless adjusted by Owner in writing, Engineer will perform the following Services:

MECHANICAL, ENGINEERING, PLUMBING, AND FIRE PROTECTION SERVICES

I. GENERAL The Mechanical Engineering, Plumbing and Fire Protection ("MEPFP") scope of work and schedule for 1-40 F Temple will align with the project scope and schedule identified in the primary Engineer proposal.

II. BASIS OF DESIGN SYSTEMS

A. CSI DIVISION 21 SYSTEMS - FIRE PROTECTION

1. NFPA 13 Full Design for Sprinkler and Fire Standpipe System (if required)
2. Documentation of Fire Pump System (as required)

B. CSI DIVISION 22 SYSTEMS - PLUMBING

1. Domestic Cold-Water System
2. Domestic Hot Water System
3. Sanitary Sewer System
4. Storm Sewer System

C. CSI DIVISION 23 SYSTEMS - HEATING, VENTILATING AND AIR CONDITIONING

1. Direct Expansion (DX) and/or Chilled Water System HVAC equipment
2. Hydronic Heating System
3. Supply Air Fan and Ductwork Systems
4. Return Air Fan and Ductwork Systems
5. Exhaust Air Fan and Ductwork Systems
6. Outside Air Systems
7. Natural Gas System
8. Energy Management & Temperature / System Control
 - Digital Temperature Monitoring and Control
 - Air Distribution System Controls
 - Hydronic System Controls

D. CSI DIVISION 26 SYSTEMS - ELECTRICAL

1. Building Power Distribution
2. Building Power Grounding Systems
3. Emergency Power Distribution and Generator (if necessary), including Life Safety and Equipment Branch
4. General and Emergency Egress Lighting – refer to assumptions and clarifications
5. Power system and raceway design to support security, audio/visual and telephone/data systems – refer to assumptions and clarifications

E. CSI DIVISION 27 SYSTEMS – TELECOMMUNICATIONS

1. Refer to assumptions and clarifications.

F. CSI DIVISION 28 SYSTEMS – SAFETY AND SECURITY

1. Fire alarm systems: NFPA 72 Full Design for Fire Alarm System and Annunciation
2. Refer to assumptions and clarifications

III. SERVICES AND DELIVERABLES

A. GENERAL - Engineer will be provided with any Owner prototype design standards or policies that will impact this scope prior to project initiation.

B. PHASE 2 - DESIGN DEVELOPMENT

Engineer will develop drawings based on the approved architectural plans and the MEPFP requirements, criteria, and system concepts developed and approved during the Schematic Design phase.

1. Services:
 - a. Attend coordination meeting weekly as per schedule.

- b. Review and evaluate updated architectural documents, structural documents, other consultants' documents, design objectives, and design and operating requirements.
 - c. Review and finalize design criteria and program requirements.
 - d. Coordinate MEP/FP space and louver requirements.
 - e. Coordinate with Architect, Structural Engineer, and other consultants on equipment supports, weights, slab openings, shaft requirements, and lateral restraints.
 - f. Provide gas load estimates and assist architect and owner in gas utility coordination as needed.
 - g. Provide electrical load estimates and assist Architect and owner in utility coordination.
 - h. Prepare Design Development drawings, including riser/flow/single line diagrams; equipment block layouts with preliminary sections, primary and typical secondary distribution; detailed equipment rooms; and preliminary equipment schedules.
 - i. Preparation of outline specifications or full specifications is excluded. An anticipated specification Table of Contents (TOC) can be supplied with DD deliverable.
 - j. Provide feedback on MEPFP project construction budgets developed by others who will have cost-estimating and cost-control responsibility throughout the entire project.
2. Deliverables
- a. Update Basis of Design to capture final design criteria.
 - b. Issue one 50% coordination document set and one 100% design development documentation set. Design Development drawings include riser/flow/single line diagrams, central plant equipment layouts with preliminary sections, local equipment room block layouts, primary and typical secondary distribution, preliminary equipment schedules, and preliminary design and construction details – BIM LOD 200
 - c. Draft of Book MEP/FP Specifications table of contents.

C. PHASE 3 – ZONING

- 1. Support zoning up to 3 zoning meetings to respond to questions and address minor revisions.

D. PHASE 4 - CONSTRUCTION DOCUMENTS, PERMITTING & BIDDING

Engineer will prepare construction drawings and specifications suitable for competitive bidding and filing with AHJ for permitting. The final design will be based on the Design Development drawings approved by Architect and Owner.

- 1. Services:
 - a. Review and evaluate final architectural documents, coordinate with structural and other consultants' documents, to determine equipment, design, and operating requirements for incorporation into final construction documents.
 - b. Finalize designs for:
 - Utility Services
 - HVAC Equipment
 - Normal Power
 - Heating and Air Conditioning Distribution
 - Supplemental Cooling
 - Ventilation, Exhaust, and Filtration
 - Electrical power distribution and Emergency/stand by power supply
 - Circuiting of internal and external building lighting systems
 - Life Safety/Fire Protection (performance specification only)
 - Lightning Protection System (performance specification only) where needed.
 - Plumbing
 - Building Management System
 - Energy Code Compliance path
 - c. Confirm final MEPFP space and louver requirements, structural requirements, and ceiling coordination with the project team.
 - d. Prepare coordinated Mechanical, Electrical, Plumbing, and Fire Protection design documents appropriate for obtaining and reviewing contractor bids. Design documents include drawings and full specifications using owner required RIB SpecLink.

- e. The fire protection system design will include performance specifications and location of riser, protection zones, architecturally coordinated heads' placement and standpipe/fire pump locations, for bidding by qualified design-build fire protection contractors that will become the engineer of-record.
 - f. The fire alarm system design will include full design, for bidding by qualified design-build fire alarm contractors that will become the engineer of- record.
 - g. Prepare Signed-and-sealed drawings and forms as necessary for filing with the AHJ (all filing to be done by others).
 - h. Review project budgets for MEPFP construction developed by others who will have cost-estimating and cost-control responsibility throughout the entire project.
 - i. Prepare final load calculations for the project systems.
 - j. Coordination with trades in-house and with Architect.
 - k. Unlimited virtual coordination Teams calls.
 - l. Respond to questions from contractors during the bidding and negotiation period.
 - m. Respond to AHJ review comments and issue clarifications where necessary.
2. Deliverables:
- a. Issue one 50% coordination document set and one 100% construction document set which will be issued for bidding, construction, and filing. Construction Documents include riser/flow/single line diagrams, central plant plans, elevations, and details, mechanical rooms and floor plans, equipment and panel schedules, and installation and construction details.
 - b. Final MEP/FP Book Specifications will be written on Owner's SpecLink.
 - c. Utility application for gas, electric and water service where necessary.
 - d. Sign and seal drawings and forms as necessary for filing with the Building Department. All forms to be populated, prepared and all filing to be done by others.

F. PHASE 5 - CONSTRUCTION ADMINISTRATION

1. Services:
- a. Review shop drawings and submittals required by contract documents. Review of contractor submittals is limited to two (2) consecutive reviews for any single submission. Additional reviews will be considered additional services unless issue is with designers' drawings. The length of the submittal review period will not exceed ten (10) business days.
 - b. Respond to any Requests for Information (RFIs) to clarify the intent of the Contract Documents. It will be the obligation of the Construction Manager to review all RFIs submitted from sub- contractors and only forward to the design team those questions with answers that are not readily apparent or inferable from the Contract Documents. RFI responses will be within 10 working days.
 - c. Review written contractor requests for change orders.
 - d. Attend a maximum of ten (10) project OAC meetings during active MEP construction and perform a maximum of ten (10) site observations including the preparation and issuance of an ongoing punch list of Mechanical and Electrical deficiencies as construction progresses through completion.
 - e. Prepare a final punch-list upon Substantial Completion of the work, prior to occupancy, including Project Sign-off (random spot checking) after the Construction Manager has certified that all corrective work has been successfully completed.
1. Deliverable:
- a. Issue on consolidated record BIM/Drawing set capturing changes made during the construction process. This will be issued at the end of construction.

STRUCTURAL ENGINEERING SERVICES

I. GENERAL

The Structural Engineering scope of work and schedule for 1-40 F Temple will align with the project scope and schedule identified in the primary Engineer proposal.

Engineer will be provided with any LDS prototype design standards or policies that will impact this scope prior to project initiation. The temple will be designed based on a prototype model provided by Owner, but significant site adaptations are assumed to be required to engineer the structure.

II. BASE OF SCOPE SERVICES

A. AREAS OF SERVICE

The general scope of work for the project includes full-service Structural design services for the building elements of Block B:

- Foundations
- Substructure
- Floor framing
- Roof framing
- Columns
- Shear walls
- Equipment support framing and grillages
- Exterior framing for stone skin

B. PHASE 2 - DESIGN DEVELOPMENT

a. Services:

- i. Attend weekly meetings based on schedule.
- ii. Finalize the structural design criteria and incorporate geotechnical consultants' foundation recommendations.
- iii. Perform more detailed analyses and further develop the structural design in coordination with Architectural and other consultants' design requirements.
- iv. Prepare Design Development level documentation defining all major structural systems and elements and including floor plans, building sections, key details, sections, and selected key member sizes. Documents will be suitable for GMP pricing.
- v. Participate in Value Engineering exercises and other cost control efforts.

b. Deliverables:

- i. Update Basis of Design document as needed.
- ii. Issue one (1) 50% coordination document set and one (1) 100% design development document set
- iii. Prepare draft structural book specification table of contents.

C. PHASE 3 – USE AND ZONING APPROVAL

- a. Services: support zoning up to 3 zoning meetings to respond to questions and address minor revisions.

D. PHASE 4 - CONSTRUCTION DOCUMENTS, PERMITTING & BIDDING

a. Services:

- i. Attend bimonthly coordination meetings
- ii. Finalize the structural design and complete the Structural Drawings and Specifications for the primary structural system in coordination with Architectural and other consultant's designs.
- iii. Provide design criteria for delegated elements to be designed by specialty engineers. Provide design criteria for pre-engineered structural elements.
- iv. Review effect of secondary and non-structural elements attached to primary structural system.
- v. Establish testing and inspection requirements.
- vi. Assist in evaluating bidders' qualifications and answer contractor questions during the bidding period.

Assist in pre-bid conference(s) with prospective bidders as needed.

b. Deliverables:

- i. Issue one (1) 50% coordination document set and one (1) 100% construction documentation set, stamped and sealed to be used for bidding, construction and filing purposes.
- ii. Provide structural book specifications using owner required RIB SpecLink.

E. PHASE 5 - CONSTRUCTION ADMINISTRATION

a. Services:

- i. Review shop drawings and submittals designed by Engineer and required by contract documents. Review of contractor submittals is limited to two (2) consecutive reviews for any single submission. Additional reviews will be considered additional services unless issue is with designer drawings. The length of the submittal review period will not exceed ten (10) business days
- ii. Review submittals for delegated design and pre-engineered structural elements for conformance with prescribed performance criteria.
- iii. Assist in selection of testing agency and establishing procedures for testing and inspections and review testing and inspection reports and recommend appropriate action as necessary. Design services as required to resolve or repair contractor deficiencies are not included but can be provided for an additional fee.
- iv. Respond to any Requests for Information (RFIs) to clarify the intent of the Contract Documents. It will be the obligation of the Construction Manager to review all RFIs submitted from sub- contractors and only forward to the design team those questions with answers that are not readily apparent or inferable from the Contract Documents. RFI responses will be within 10 working days.
- v. Review written contractor requests for change orders.
- vi. Attend a maximum of ten (10) project OAC meetings during active MEP construction and perform a maximum of ten (10) site observations including the preparation and issuance of an ongoing punch list of Mechanical and Electrical deficiencies as construction progresses through completion.
- vii. Prepare a final punch-list upon Substantial Completion of the work, prior to occupancy, including Project Sign-off (random spot checking) after the Construction Manager has certified that all corrective work has been successfully completed.

b. Deliverable:

- i. Issue on consolidated record BIM/Drawing set capturing changes made during the construction process. This will be issued at the end of construction.

ARCHITECTURAL LIGHTING SERVICES

I. GENERAL

- A. The Architectural Lighting scope of work and schedule for 1-40 F Temple will align with the project scope and schedule identified in the primary Engineer proposal.
- B. The general scope of work for the project includes lighting design services for the following areas:
 1. Building Exterior: Exterior architectural feature lighting, Entrance areas immediately adjacent to building, Porte Cocheres, Entry Plaza, Parking and Pedestrian Site Lighting. 3D Photometric will be provided.
 2. Interior Areas: Main lobby entrance, public restrooms, Instruction Rooms, Admin and Waiting Areas, Dressing Rooms for patrons and workers, Baptistry, Celestial Room, Sealing Rooms, 3D photometric for these rooms, Elevator lobbies, Lighting design for all areas of the Temple.

II. BASE DESIGN SERVICES

A. PHASE 2 – DESIGN DEVELOPMENT

1. Services

- a. Attend workshop meetings with Owner, Architect, and other consultants to establish aesthetic criteria, illumination criteria, sustainability criteria, project constraints (budget, schedule, etc.) and coordinate design concepts.

- b. Review of Owner furnished Standard Plan, Temple Program, Temple Design Guidelines and Core Details, and other Owner Project Requirements.
- c. Participate in team conference calls or web-based meetings as required. Time for up to four (4) hours of teleconference time attended by the lead lighting designer is included in the Fee.

2. Deliverables:

- a. Prepare a Basis of Design Lighting Narrative that describes the design concepts, criteria, standards and lighting design intent with sketch layouts, light level goals and precedent images for review. Approved concepts will be the basis for continued work in Design Development.
- b. This phase includes one deliverable: One (1) SD phase narrative.

B. PHASE 3 – ZONING AND USE (NA)

- 1. Support up to 2 zoning meetings to respond to questions and address minor revisions.

C. PHASE 4 - CONSTRUCTION DOCUMENTATION AND BIDDING PHASE

1. Services:

- a. Attend workshop meetings to finalize and coordinate the lighting design.
- b. Participate in team conference calls or web-based meetings as required. Time for up to eight (8) hours of teleconference time attended by the lead lighting designer is included in the Fee.
- c. Final coordination of decorative light fixtures specified by the Interior Designer to verify wattage, lumen output, CRI and color temperature.
- d. Review and comment on architectural and electrical lighting plans to check that lighting information has been incorporated accurately. Time for two (2) reviews at 50% CDs and 95% CDs is included in the Fee.
- e. Final coordination of installation details illustrating design intent, proposed mounting, and dimensional requirements, etc. Architect will be responsible for final construction details based on provided sketches.

2. Deliverables:

- a. CSI specifications for the architectural lighting and final luminaire schedule to be included in the electrical construction documents using Owner's required RIB SpecLink.
- b. Control intent plan diagrams showing control zones and control device locations, and control intent schedule indicating sequence of operations for incorporation into the electrical engineering lighting control document.
- c. Final luminaire schedule, fixture cuts and sketches as required to explain the proposed lighting design.
- d. ComCheck lighting input.
- e. This phase includes two deliverables: One (1) 50% CD set, and One (1) 100% CD IFC set.

D. PHASE 5 - CONSTRUCTION ADMINISTRATION- Submittals and Shop Drawings Review

1. Services:

- a. Provide review and documentation of comments for submittals, shop drawings and required samples and/or submissions of specified lighting fixtures and controls. Architect must also review lighting shop drawings. We allow for a maximum of two (2) submittal packages.
- b. Provide consultation, as necessary, to clarify and respond to Contractor's "Requests for Information". Time for review is included in the Fee.
- c. A maximum of one (1) 3-day field visit is included for fixture aiming, including inspection of the final lighting installation. Engineer will supervise the focusing of adjustable fixtures. The Contractor is to provide personnel to perform the final focusing under the direction of Engineer Lighting Department.

- d. During the final inspection, Engineer Lighting will provide personnel for a maximum one (1) eight- hour period (day, dusk, night) to supervise setting of light levels for the dimming system. Manufacturer of dimming system will provide personnel to program and commission the dimming system. Additionally, Owner's representative will be available and present at commissioning for training and "handover" of system.

2. Deliverables:

- a. Provide final written document indicating field observations and punch list items.

BUILDING ENCLOSURES CONSULTING SERVICES

III. GENERAL

- A. The Building Enclosure scope of work and schedule for 1-40 F Temple will align with the project scope and schedule identified in the primary Engineer proposal.
- B. Architect will provide Engineer all progress CSI-format specifications for relevant systems, finishes, and devices during the period of Engineer services, in order for Engineer to provide analysis reports and conduct design coordination in a timely and thorough fashion.
- C. Architect will incorporate the work of all specialty consultants, such as interior design consultant into the architectural Revit model.
- D. Engineer will provide written comments and mark up on base drawings and technical specifications provided by Architect for incorporation into building enclosure elements documentation at each submission of the design phase.

IV. BASE DESIGN SERVICES

A. PHASE 2 - DESIGN DEVELOPMENT (DD)

Input at this stage is typically used by the design team to compare system detail options for cladding systems, air and water barriers, flashing, fenestration, and roofing details. This phase includes:

1. Services:

- a. Attend working sessions to discuss interface details for cladding systems, air and water barriers, roofing and waterproofing, and to refine flashing details.
- b. Review specifications and comment on requirements for roofing, air barrier, waterproofing, exterior cladding systems, fenestration, and recommend testing and inspections. Engineer will provide electronic mark-ups directly on the PDF specifications with comments and sample specifications during above mentioned working sessions.
- c. Complete hygrothermal analyses of 3 wall types to assess moisture movement in wall and roof assemblies, the potential for condensation and moisture accumulation, and to recommend the location of the vapor control layer using WUFI, a 1-D heat and moisture transfer modeling program, to complete the analyses. To avoid reorganization of wall materials, we recommend performing hygrothermal analyses early in the design phase.
- d. Provide detail concepts or hand sketches for the designer to review and incorporate.
- e. Review 100% Design Development drawings and provide written comments.
- f. Attend one meeting with the design team to discuss Engineer's comments on 100% Design Development drawings.
- g. Conduct Design Development (DD) review and meeting for masonry walls, window wall systems, and/or individual window systems.
 - One (1) workshop with Architect.
 - Provide up to 7 concept details for the wall types as part of Engineer's mark-ups of Architect's drawings. This includes typical and interface conditions.
 - Review specifications for the glazed wall types and comment on requirements for curtain wall and

recommending testing and inspections.

- Review and comment on interfaces between the walls and adjacent materials.
- Refined sizing of face glass and framing members.
- Review building and wall movements and tolerance for consideration in the design.
- Work with building engineer for anchorages and loads.
- Work with building mechanical engineer for wall criteria.
- Recommend scope of site testing and any mockups (visual, in-place, and performance mockup at a laboratory is not expected to be recommended).

2. Deliverables:

- a. Review and comment on the 100% DD drawings for the building enclosure and curtain wall provided by Architect as a marked-up PDF set.
- b. Provide one (1) review of the draft CSI format specification with written comments.

B. PHASE 3- USE AND ZONING (NOT APPLICABLE)

C. PHASE 4 - CONSTRUCTION DOCUMENTATION (CD) AND BIDDING PHASE

Engineer will review the drawings for conditions affecting thermal, air, and moisture control, and the appropriateness of the selected materials in the expected environments for the building enclosure. The scope will be completed through a mix of peer reviews and working sessions with Owner's staff.

1. Services:

- a. Review the 90% CD drawings and related technical specification sections, focused on specific assembly details and unique or complex project details. Engineer will provide electronic mark-ups directly on the PDF drawings and specifications with Engineer's comments.
- b. Attend one working 4-hour session with the design team/client to further discuss and finalize interface details for window wall and/or window system and cladding systems, air and water barriers, and roofing and waterproofing. Engineer will finalize flashing details.
- c. Engineer will review the HVAC system interaction with the building enclosure but will not complete a full mechanical system review.
- d. Conduct Construction Documentation (CD) review and meeting for masonry walls, window wall systems, and/or individual window systems.
 - One (1) half-day workshop with architect.
 - Refine details for the curtain wall as part of Engineer's mark-ups of Architect's drawings. This includes typical and interface conditions.
 - Review specifications for the glazed wall types and comment on requirements for curtain wall at the CD phase.
 - Review building and wall movements and tolerance for consideration in the design.
 - Work with building engineer for refined anchorages and loads.
 - Work with building mechanical engineer for any changes to the wall criteria.
 - Review functional performance testing criteria and recommend changes.
 - Review and comment on the 100% CD drawings for the curtain wall.
- e. Attend one pre-bid conference to answer questions from prospective bidders.
- f. Review and comment on up to four (4) sub-contractor proposals from the various trades.
- g. Review and comment on bidders' value engineering proposals, up to 4 requests.

2. Deliverables:

- a. Review the 90% Construction Document drawings and technical specification sections, focused on specific assembly details and unique or complex project details. Provide electronic mark-ups directly on the PDF drawings and specifications with Engineer's comments.

- b. Review and provide comments on bidding documents, including subcontractor proposals and value engineering proposals, up to the quantity specified above.

D. PHASE 5 - CONSTRUCTION ADMINISTRATION- Submittals and Shop Drawings Review

1. Services:

- a. Conduct up to five (5) reviews of enclosure system shop drawings and corresponding material submittals for the following systems:
 - Below grade and at grade waterproofing (one system).
 - Window wall, punched windows and / or curtain wall (up to two systems).
 - Roofing.
 - One wall rainscreen or opaque wall system.
- b. Engineer will provide comments via mark-ups of the drawing sets and provide a summary memo identifying major concerns.
- c. Review of contractor submittals is limited to two (2) consecutive reviews for any single submission. Additional reviews will be considered additional services. The length of the submittal review period will not exceed ten (10) business days.
- d. Conduct up to three (3) site visits. These visits will be performed by local Engineer staff trained and directed by the Building Enclosures team. After each visit, we will provide a site visit report with representative photographs to illustrate findings with a summary list of recommended corrective action that will need to be completed. Some of these visits can be utilized to observe functional performance testing completed by others for air and water infiltration and sealant adhesion.

ACOUSTICS SERVICES

I. GENERAL

The Acoustics, Noise and Vibration Control scope of work and schedule for 1-40 Temple will align with the project scope and schedule identified in the primary Engineer proposal.

A. GENERAL

1. Refer to Owner prototype standards for definition of each deliverable.
2. Interior Sound Isolation - Performance criteria and detailed recommendations to control sound transmission between interior programmed spaces.
3. Interior Room Acoustics - Design guidance on a per-space basis, focusing on interior finish selections and architectural finishes in Lobbies and Amenity Spaces (i.e. meetinghouse).
4. MEP Systems Noise & Vibration Control- Performance criteria and detailed recommendations to control MEP system noise and vibration in all interior occupied spaces, and to meet all local noise ordinances at the property line.

B. PHASE 2 – FLIGHTING

C. DESIGN DEVELOPMENT

1. Services:

- a. Attend up to twelve (12) virtual meetings for Acoustical coordination. Attendance on one (1) weekly virtual meeting is assumed. If schedule extends and additional meetings are required, additional services would be required.
- b. Review schematic design and programming requirements to develop recommended acoustical performance criteria for each interior space.
- c. Provide guidance on sound isolation to meet the established STC/IIC ratings at all adjacencies. Recommendations will be provided for:
 - i. Wall partition assembly designs.
 - ii. Junction detailing.
 - iii. Floor and ceiling assembly designs.
 - iv. Door construction types, door seals, and framing.

- v. Interior and exterior glazing and window design.
 - vi. MEP equipment room / enclosure designs (for mitigation of equipment noise transmission to adjacent spaces).
- d. Identify suitable acoustic finishes to control reverberation and meet the established interior acoustical performance criteria.
 - e. Coordinate with Architect regarding recommendations for exterior glazing to reduce noise ingress from the exterior and to meet the STC design criteria.
 - i. An exterior noise assessment is not included in the base scope. In lieu of measurements at the project site, Engineer will provide recommendations to meet LDS façade design criteria and will review site surroundings to determine if a higher level of sound isolation would be recommended.
 - f. Coordinate with MEP engineers to select equipment to achieve the established background noise criteria. Recommendations will be provided regarding:
 - i. Sizing of mechanical equipment.
 - ii. Preliminary locations and performance criteria for sound attenuators, duct lining, external duct wrap, vibration isolators and equipment mounts, as needed.

2. Deliverables:

- a. Prepare and issue one (1) Design Development Acoustical Report which documents all of the design recommendations and considerations listed above. This will be part of the Basis of Design document. The report will include:
 - i. Design criteria for each programmed space type.
 - ii. Partition types and floor plan mark-ups outlining the recommended strategy to achieve required STC ratings.
 - iii. Structural requirements and/or preliminary floor-ceiling construction recommendations to achieve the required STC and IIC ratings.
 - iv. Room shaping & finish selection strategies to achieve recommended interior room acoustical criteria.
 - v. Template criteria for controlling noise & vibration from building systems.

E. PHASE 3 – USE AND ZONING (NOT APPLICABLE)

F. PHASE 4 - CONSTRUCTION DOCUMENTS - REVIEW AND MARK UP

1. Services:

- a. Attend up to twenty (20) virtual meetings for Acoustical coordination. Attendance on one (1) weekly virtual meeting is assumed. If schedule extends and additional meetings are required, additional services would be required.
- b. Review 100% Design Development documents to confirm compliance with acoustical criteria and recommendations in the Design Development Acoustical Report. An additional review of a Construction Documents progress set (ex: 70% CD) is expected.
- c. Provide guidance on all architectural elements relevant to sound transmission between spaces in the form of sketches, markups, and written recommendations.
- d. Review finishes and identify alternate options in order to meet the established interior acoustical performance criteria.
- e. Further coordinate with MEP engineers to inform the system designs ability to achieve the established background noise criteria. Written recommendations, sketches, and markups will be provided regarding:
 - i. Selection and specification of sound attenuators, acoustical louvers, and internal/external duct lining or lagging.
 - ii. Selection and specification of vibration isolators, curbs, and duct/piping mounts.
 - iii. Locations and performance criteria for all registers, diffusers, grilles, and other in-room system components.

2. Deliverables:

- a. Prepare and issue up to two (2) Construction Documents Acoustical Reports which summarize design recommendations and considerations listed above. This will be part of an update to the Basis of Design document issued with the:
 - i. 70% CD (or similar milestone) Basis of Design document and
 - ii. 100% CD Basis of Design document.

E. PHASE 5 - CONSTRUCTION ADMINISTRATION

1. Services:

- a. Answer questions of contractors during the bidding period.
- b. Review requested submittals for architectural and MEP devices and components, to confirm compliance with acoustical design criteria. Up to two (2) reviews of submittal reissuances are included.
- c. Review and respond to requests for information concerning the acoustical design, as well as coordination and installation of acoustical finishes and devices.

F. SITE VISITS RECOMMENDED

1. Services:

- a. Conduct up to six (6) virtual site assessments during construction to verify proper installation of acoustical finishes and devices. Field observation trips assumed to be virtual visits assisted by another Engineer discipline onsite. If Engineer is not selected for site visit services by other disciplines, these visits will need to be assisted by Architect.
- b. Prepare and issue a field observation report of any deficiencies observed during the site assessment(s) requiring remedial action by Contractor.

CIVIL ENGINEERING SERVICES

I. GENERAL

The Civil Engineering scope of work and schedule for the 1-40 F Temple will align with the project scope and schedule identified in the primary Engineer proposal. The Civil Engineering services will cover the site development, utility connections, stormwater management, and other related civil engineering items as required for the temple and associated buildings.

II. BASE OF SCOPE SERVICES

A. AREAS OF SERVICE

The general scope of work for the project includes full-service Civil Engineering design services for the following elements of the site:

1. Site grading and earthwork.
2. Stormwater management systems.
3. Utility connections (water, sewer, natural gas, power, internet, etc.).
4. Site access, including roads, driveways, and parking layouts.
5. Pedestrian pathways and circulation.
6. Erosion and sediment control.
7. Off-site improvements as necessary for site integration.

B. PHASE 2 - DESIGN DEVELOPMENT

1. Services:

- a. Attend weekly meetings based on the project schedule.
- b. Finalize the civil design criteria, incorporating inputs from geotechnical, environmental, and other relevant studies.
- c. Perform detailed analyses of the site, including stormwater management, and develop the design in coordination with architectural and landscape designs.
- d. Prepare Design Development level documentation defining all major civil systems, including grading

plans, stormwater management strategies, utility layouts, and paving plans. These documents will be suitable for GMP pricing.

- e. Participate in Value Engineering exercises and other cost control efforts.

2. Deliverables:

- a. Update Basis of Design document as needed. Prepare site plan drawings to inform the design including:
 - i. Existing Conditions & Removals Plan
 - ii. Refined Overall Site Plan
 - iii. Refined Mass Grading Plan.
 - iv. Detailed Site Grading Plan
 - v. ADA Driveway/Ramps Details
 - vi. Detailed Overall Utility Plan.
 - vii. Sanitary Sewer Plan
 - viii. Storm Drainage
 - ix. Erosion Control Plan
 - x. Construction Details
- b. Civil Site Schematic Design Cost Estimate.
- c. Issue one (1) 50% coordination document set and one (1) 100% design development document set.
- d. Prepare draft civil specifications, including materials and construction methods.

C. PHASE 3 – USE AND ZONING APPROVAL

1. Services:

- a. Support zoning approval processes by preparing necessary documentation and attending up to three (3) zoning meetings to respond to questions and address minor revisions.

D. PHASE 4 - CONSTRUCTION DOCUMENTS, PERMITTING & BIDDING

1. Services:

- a. Attend bi-monthly coordination meetings.
- b. Finalize the civil design and complete the Civil Engineering Drawings and Specifications in coordination with architectural and other consultants' designs.
- c. Establish testing and inspection requirements for civil works.
- d. Assist in evaluating bidders' qualifications and answer contractor questions during the bidding period.
- e. Assist in pre-bid conference(s) with prospective bidders as needed.

2. Deliverables:

- a. Issue one (1) 50% coordination document set and one (1) 100% construction documentation set, stamped and sealed to be used for bidding, construction and filing purposes.
- b. Provide full civil specifications will be written on owner SpecLink.

E. PHASE 5 - CONSTRUCTION ADMINISTRATION

1. Services:

- a. Review shop drawings and submittals required by the contract documents.
- b. Provide on-site inspections to ensure civil works conform to the contract documents.
- c. Respond to RFIs related to civil engineering aspects within ten (10) business days.
- d. Review written contractor requests for change orders related to civil works.

- e. Conduct a maximum of ten (10) site observations during active construction and prepare ongoing punch lists for civil works.
 - f. Prepare a final punch-list upon Substantial Completion, including site sign-off after Construction Manager certification of completion.
2. Deliverable:
- a. Issue a consolidated record drawing set capturing changes made during the construction process, to be issued at the end of construction.